UC SAN DIEGO / ECE 251C

2

Continuous-domain signal processing

Executive summary

Continuous-domain filtering follows the same principles as discrete-domain filtering, with integrals replacing sums. The Dirac impulse makes this correspondence precise through its sifting property. We use it to describe convolution, distinguish finite and infinite impulse responses through support, and relate the Fourier and Laplace transforms. The centered box filter is a moving average whose frequency response is a sinc. Derivatives of the impulse give a second class of examples: convolution with $\delta^{(n)}$ differentiates a signal n times and has frequency response $(\mathrm{j}\omega)^n$.

2.1 Signals and systems on the real line

A discrete-domain signal assigns a value to each integer index. A continuous-domain signal f instead assigns a value $f(t)$ to each $t \in \mathbb{R}$. The coordinate t need not represent time: it may describe position along a line. In higher dimensions, the domain may describe location in an image; here we work on $\mathbb{R}$. The word *continuous* refers to the domain; the function itself may have jumps. As before, a system H maps the input signal f to an output signal $g = \mathrm{H}f$.

Linearity. If $g_1 = \mathrm{H}f_1$ and $g_2 = \mathrm{H}f_2$, then

$$\mathrm{H}(\alpha f_1 + \beta f_2) = \alpha g_1 + \beta g_2 \qquad (\alpha, \beta \in \mathbb{R}). \tag{2.1}$$

Translation invariance. Define $(T_a f)(t) \coloneqq f(t-a)$. When t represents time, a positive a is a delay and we call this *time invariance*. In either interpretation,

$$\mathrm{H}(T_a f) = T_a(\mathrm{H}f) \qquad \text{for every } a \in \mathbb{R}. \tag{2.2}$$

Unlike the discrete case, the shift a may be *any real number*.

2.2 The Dirac impulse: unit mass at one point

The Kronecker impulse is nonzero at exactly one index, and its sum is one. An ordinary function that is nonzero at only one point has integral zero, regardless of its finite value at that point. We therefore cannot obtain a continuous-domain

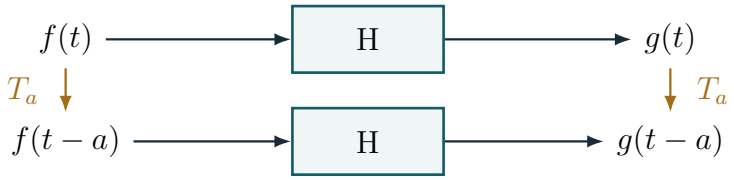


Figure 2.1: Translation invariance: shifting the input shifts the output by the same amount. For a temporal signal, the shift is a time shift.

unit impulse by simply replacing the integer index in the Kronecker definition with a real variable.

Instead, we define the *Dirac impulse* by what it does when integrated against a signal. For a function φ that is continuous at a,

$$\boxed{\int_{\mathbb{R}} \varphi(t)\delta(t-a)\,\mathrm{d}t = \varphi(a).} \tag{2.3}$$

Equation (2.3) is the *sifting property*: integration against an impulse at a returns the value of the function at a. This action defines the impulse. In particular, the displayed integral should not be interpreted as an ordinary integral involving a pointwise value $\delta(a)$. The appropriate interpretation is a unit point mass concentrated at a.

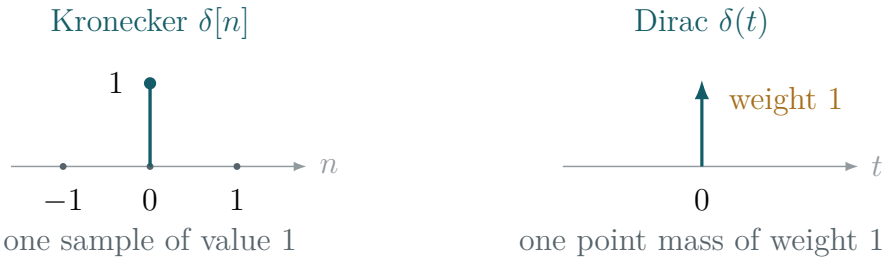


Figure 2.2: The arrow for a Dirac impulse records its location and weight. Its drawn height is not the value of $\delta(0)$.

A limiting picture. For $\varepsilon > 0$, let $r_\varepsilon(t) = \varepsilon^{-1}$ on $[-\varepsilon/2, \varepsilon/2]$ and zero elsewhere. Its width is ε, its height is $1/\varepsilon$, and its area is one. Then

$$\begin{aligned}\left|\int_{\mathbb{R}} r_\varepsilon(t)\varphi(t)\,\mathrm{d}t - \varphi(0)\right| &\le \frac{1}{\varepsilon}\int_{-\varepsilon/2}^{\varepsilon/2} |\varphi(t)-\varphi(0)|\,\mathrm{d}t \\ &\le \sup_{|t|\le\varepsilon/2} |\varphi(t)-\varphi(0)| \;\longrightarrow\; 0.\end{aligned}$$

Continuity at zero makes the last supremum tend to zero. Thus the narrowing pulses reproduce the defining action of δ after integration against any function continuous at zero. The convergence concerns these integrals, not pointwise convergence to an ordinary function with unit area.

2.3 Three consequences of sifting

Each identity below follows from the defining action; none requires assigning a pointwise value to δ.

1. The impulse has total mass one. Use the constant function $\varphi(t) = 1$:

$$\int_{\mathbb{R}} \delta(t)\,\mathrm{d}t = \int_{\mathbb{R}} 1 \cdot \delta(t)\,\mathrm{d}t = 1. \tag{2.4}$$

More generally, $c\delta(t-a)$ has weight c and evaluates φ as $c\varphi(a)$. A negative weight is allowed; it means subtraction, not a negative time location.

2. Its Fourier transform is constant. We use angular frequency ω and the convention

$$\widehat{f}(\omega) \coloneqq \int_{\mathbb{R}} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t, \qquad \mathrm{j}^2 = -1. \tag{2.5}$$

For an ordinary signal, absolute integrability is a sufficient condition for this integral. For the impulse, the integral is evaluated by sifting:

$$\widehat{\delta}(\omega) = \int_{\mathbb{R}} \delta(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t = \mathrm{e}^{-\mathrm{j}\omega\cdot 0} = 1. \tag{2.6}$$

The same calculation at a gives $\widehat{\delta(\cdot - a)}(\omega) = \mathrm{e}^{-\mathrm{j}\omega a}$. A delay changes the phase, but not the magnitude, of this spectrum.

3. Convolution with an impulse leaves the signal unchanged. Continuous-time convolution is

$$(h * f)(t) \coloneqq \int_{\mathbb{R}} h(u) f(t-u)\,\mathrm{d}u$$

whenever the integral is defined. For an impulse kernel, apply sifting to the function $u \mapsto f(t-u)$:

$$(\delta * f)(t) = \int_{\mathbb{R}} \delta(u) f(t-u)\,\mathrm{d}u = f(t). \tag{2.7}$$

Likewise, a shifted impulse gives $(\delta(\cdot - a) * f)(t) = f(t-a)$. Thus δ is the impulse response of the identity system, and $\delta(\cdot - a)$ is the impulse response of a pure delay.

> **One definition, three interpretations**
>
> The Dirac impulse has unit mass, a flat Fourier spectrum, and the identity action under convolution. These are different consequences of the same sifting rule.

2.4 From impulse response to convolution

The *impulse response* of a continuous-time system is $h := \mathrm{H}\delta$. Time invariance specifies the response to an impulse at any other location:

$$\mathrm{H}\{\delta(\cdot - u)\}(t) = h(t - u).$$

Linearity then gives the response to a finite collection of weighted impulses. For example,

$$\mathrm{H}\{2\delta + \delta(\cdot - a)\}(t) = 2h(t) + h(t - a).$$

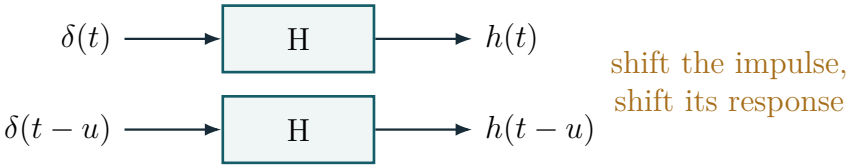


Figure 2.3: Every translated impulse produces a translated copy of the same response.

The same sifting rule expresses an input as a continuous superposition of shifted impulses. Indeed, fixing t and integrating with respect to u gives

$$f(t) = \int_{\mathbb{R}} f(u)\delta(t - u)\,\mathrm{d}u. \tag{2.8}$$

Assuming the system respects this integral superposition, linearity and time invariance give

$$\begin{aligned}(\mathrm{H}f)(t) &= \int_{\mathbb{R}} f(u)\,\mathrm{H}\{\delta(\cdot - u)\}(t)\,\mathrm{d}u \\ &= \int_{\mathbb{R}} f(u)h(t - u)\,\mathrm{d}u \\ &= \int_{\mathbb{R}} h(v)f(t - v)\,\mathrm{d}v = (h * f)(t),\end{aligned} \tag{2.9}$$

where the last equality follows from $v = t - u$ and reversal of the integration limits. This is the continuous-domain convolution formula. Just as in the discrete case, each input value weights a translated copy of the impulse response; the superposition is now an integral rather than a sum.

The role of the integral. Linearity by itself concerns finite sums. The impulse-response derivation also assumes that the system respects the limit used to form the integral superposition. Alternatively, the convolution formula directly defines a linear system. The identity

$$(h * T_a f)(t) = \int_{\mathbb{R}} h(v)f(t - a - v)\,\mathrm{d}v = (h * f)(t - a)$$

proves time invariance. If $|f(t)| \le M$ and $\int_{\mathbb{R}} |h(v)|\,\mathrm{d}v < \infty$, then

$$\int_{\mathbb{R}} |h(v) f(t-v)|\,\mathrm{d}v \le M \int_{\mathbb{R}} |h(v)|\,\mathrm{d}v < \infty.$$

This is a sufficient condition for the convolution to converge absolutely.

2.5 Support, closure, and compactness

When the impulse response is an ordinary function h, we also call h the *filter*. To distinguish filters of finite and infinite duration, we first need a precise description of the region where h is nonzero. Its *support* is

$$\operatorname{supp}(h) := \overline{\{t \in \mathbb{R} : h(t) \neq 0\}}. \tag{2.10}$$

The overbar denotes *closure*. It adds any missing limit points to the set: a point belongs to the closure whenever points of the set can approach it arbitrarily closely. For example, the closure of an open interval contains its endpoints, even though the open interval does not. This convention prevents the support from depending on whether an endpoint happens to be assigned the value zero.

A set $S \subseteq \mathbb{R}$ is *closed* if it contains every limit of a convergent sequence of points in S. Its closure $\overline{S}$ includes S and all such limit points. Equivalently, $t \in \overline{S}$ if every interval $(t-\varepsilon, t+\varepsilon)$, however small, meets S.

For example, $a + (b-a)/m \in (a,b)$ for $m \ge 2$ and tends to a. Thus a belongs to the closure of (a,b) even though a does not belong to the interval itself. The same applies to b. Points outside $[a,b]$ have a small neighborhood that misses (a,b), so

$$\overline{(a,b)} = [a,b].$$



Figure 2.4: Closure adds the missing limit points. Filled endpoints are included; open endpoints are excluded.

Exercise, with answers. Assume $a < b < c < d$.

Set	Closed?	Closure
(a, b)	No	$[a, b]$
$[a, b)$ or $(a, b]$	No	$[a, b]$
$[a, b]$	Yes	$[a, b]$
$(a, b] \cup [c, d)$	No	$[a, b] \cup [c, d]$

A set is *bounded* if it lies in some finite interval $[-B, B]$. In $\mathbb{R}$, a set is *compact* precisely when it is closed and bounded. This is the real-line Heine–Borel characterization; it is the characterization we use here. The support in (2.10) is already closed, so it is compact exactly when it is bounded.

2.6 Finite and infinite impulse responses

A continuous-time filter is *FIR* if $\operatorname{supp}(h)$ is compact, and *IIR* if $\operatorname{supp}(h)$ is unbounded. Here "finite" means *finite duration*, not a finite number of real-valued time instants.

We use the indicator notation

$$\mathbf{1}_A(t) \coloneqq \begin{cases} 1, & t \in A, \\ 0, & t \notin A. \end{cases}$$

Consider the two filters

$$h_{\text{box}}(t) = \mathbf{1}_{[-1/2,\,1/2]}(t), \qquad h_{\text{tail}}(t) = \mathrm{e}^{-t}\mathbf{1}_{[0,\,\infty)}(t).$$

The centered box h_{box} is also called the *rectangular function*, often written $\operatorname{rect}(t)$. The two filters have supports $[-1/2, 1/2]$ and $[0, \infty)$, respectively. The first filter is FIR. The second is IIR: its positive tail becomes small but never becomes identically zero.

Comparison with the discrete definition. For a sequence, the support is $S = \{n \in \mathbb{Z} : h[n] \neq 0\}$. Any bounded subset of $\mathbb{Z}$ is finite: if $|n| \leq B$, only finitely many integers are possible. Conversely, every finite set is bounded. Moreover, every subset of $\mathbb{Z}$ is closed in $\mathbb{R}$: a convergent sequence of integers is eventually constant, since distinct integers are at least one unit apart. Consequently,

$$S \text{ compact} \quad \Longleftrightarrow \quad S \text{ finite}.$$

Thus the continuous-time definition has exactly the same support-based interpretation as the FIR definition for discrete filters.



Figure 2.5: A response can decay rapidly and still have infinite duration. FIR/IIR describes where the response is nonzero, not how large it is.

Duration is not causality

The centered box uses both past and future input values. It is FIR, but not causal as a real-time filter. Delaying the box by $1/2$ gives support $[0, 1]$ and a causal moving average. Compact support alone does not decide causality.

2.7 The box filter as a moving average

Let $h(t) = \mathbf{1}_{[-1/2,\,1/2]}(t)$. Substitution into convolution gives

$$\begin{aligned} g(t) = (h * f)(t) &= \int_{-1/2}^{1/2} f(t-u)\,\mathrm{d}u \\ &= \int_{t-1/2}^{t+1/2} f(v)\,\mathrm{d}v. \end{aligned} \tag{2.11}$$

For the second equality, set $v = t - u$, so $\mathrm{d}v = -\mathrm{d}u$ and the integration limits reverse. The interval has length one, so this integral is exactly the average value of f in a unit-width window centered at t.

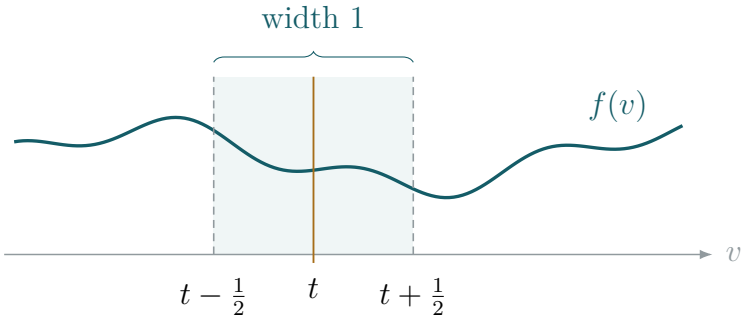


Figure 2.6: Move the window, integrate the enclosed signal, and record the result as $g(t)$. The filter is fixed; the window moves with the output time.

Worked example: a jump becomes a ramp. Let $f(t) = \mathbf{1}_{[0,\infty)}(t)$ be a unit step. The integral in (2.11) is the length of the part of $[t-1/2, t+1/2]$ lying to the right of zero. Therefore

$$g(t) = \begin{cases} 0, & t \le -\tfrac{1}{2}, \\ t + \tfrac{1}{2}, & -\tfrac{1}{2} < t < \tfrac{1}{2}, \\ 1, & t \ge \tfrac{1}{2}. \end{cases} \tag{2.12}$$

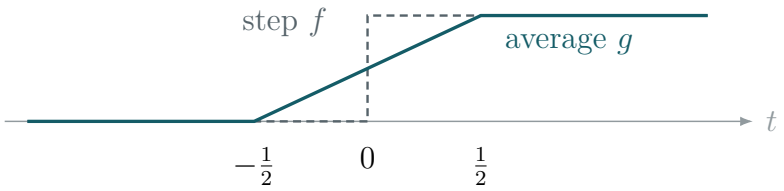


Figure 2.7: The dashed step is replaced by a linear transition of width one. Averaging preserves a constant region but spreads an abrupt change.

Averaging leaves the constant parts of the step unchanged and replaces the jump by a transition over the width of the window. This explains the smoothing directly in the signal domain. We next examine the same operation in the frequency domain.

2.8 Pure frequencies are eigenfunctions

Extend a real linear filter to complex signals by acting separately on the real and imaginary parts. Euler's formula $\mathrm{e}^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta$ then lets us study both sinusoids at once.

Suppose $\int_{\mathbb{R}} |h(u)|\,\mathrm{d}u < \infty$. For the input $f(t) = \mathrm{e}^{\mathrm{j}\omega t}$,

$$\begin{aligned} (\mathrm{H}f)(t) &= \int_{\mathbb{R}} h(u)\mathrm{e}^{\mathrm{j}\omega(t-u)}\,\mathrm{d}u \\ &= \mathrm{e}^{\mathrm{j}\omega t}\int_{\mathbb{R}} h(u)\mathrm{e}^{-\mathrm{j}\omega u}\,\mathrm{d}u \\ &= \widehat{h}(\omega)\mathrm{e}^{\mathrm{j}\omega t}. \end{aligned} \tag{2.13}$$

The integral is absolutely convergent because the exponential has magnitude one. Notice that the input itself need not have an absolutely convergent Fourier integral for this *filtering* calculation to be valid.

The function $\widehat{h}(\omega)$ is the system's *frequency response*. In engineering notation, it is also written $H(\mathrm{j}\omega)$:

$$H(\mathrm{j}\omega) = \widehat{h}(\omega) = \int_{\mathbb{R}} h(u)\mathrm{e}^{-\mathrm{j}\omega u}\,\mathrm{d}u. \tag{2.14}$$

Here H denotes the *system*, whereas $H(\mathrm{j}\omega)$ denotes a *complex number* for each frequency.

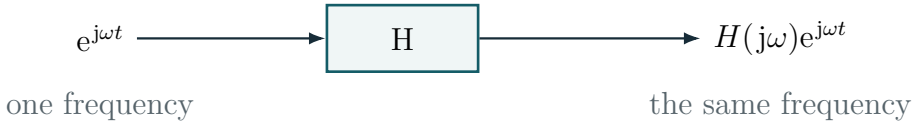


Figure 2.8: The exponential keeps its shape under an LTI filter. Its multiplier is an eigenvalue of the system.

If $H(\mathrm{j}\omega) = \rho \mathrm{e}^{\mathrm{j}\theta}$ with $\rho \geq 0$, the output is $\rho \mathrm{e}^{\mathrm{j}(\omega t+\theta)}$. For a real filter, taking real parts yields

$$\cos(\omega t) \;\longmapsto\; \rho \cos(\omega t + \theta).$$

Thus $|H(\mathrm{j}\omega)|$ describes the amplitude gain; an argument of $H(\mathrm{j}\omega)$ describes the phase shift when the response is nonzero.

Unlike discrete-time exponentials, continuous-time exponentials at ω and $\omega + 2\pi$ are generally different: $\mathrm{e}^{\mathrm{j}2\pi t}$ need not equal one for real t. The continuous-time frequency response need not be periodic.

2.9 Convolution becomes multiplication

The response to a complex exponential suggests a simpler description of an LTI system: each frequency is multiplied by the corresponding value of $\widehat{h}$. The convolution theorem makes this statement precise for general inputs. We first prove it under absolute-integrability assumptions, which justify changing the order of integration.

Proposition 2.1 (Convolution theorem)**.** Suppose that h and f are absolutely integrable:

$$\int_{\mathbb{R}} |h(t)|\,\mathrm{d}t < \infty, \qquad \int_{\mathbb{R}} |f(t)|\,\mathrm{d}t < \infty.$$

Then $g = h * f$ is defined almost everywhere (except on a set of length zero), is absolutely integrable, and

$$\widehat{g}(\omega) = \widehat{h}(\omega)\widehat{f}(\omega). \tag{2.15}$$

Equivalently, $G(\mathrm{j}\omega) = H(\mathrm{j}\omega)F(\mathrm{j}\omega)$.

Proof. First integrate the nonnegative absolute value; nonnegative integrands permit either order of integration:

$$\begin{aligned}\int_{\mathbb{R}}\int_{\mathbb{R}} |h(u)f(t-u)|\,\mathrm{d}u\,\mathrm{d}t &= \int_{\mathbb{R}} |h(u)| \left(\int_{\mathbb{R}} |f(t-u)|\,\mathrm{d}t\right) \mathrm{d}u \\ &= \left(\int_{\mathbb{R}} |h(u)|\,\mathrm{d}u\right)\left(\int_{\mathbb{R}} |f(v)|\,\mathrm{d}v\right) < \infty.\end{aligned}$$

The substitution in the second line is $v = t - u$. The same bound controls $\int_{\mathbb{R}} |g(t)|\,\mathrm{d}t$. Absolute integrability permits reversing the order of integration in the Fourier calculation:

$$\begin{aligned}\widehat{g}(\omega) &= \int_{\mathbb{R}}\int_{\mathbb{R}} h(u)f(t-u)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}u\,\mathrm{d}t \\ &= \int_{\mathbb{R}} h(u)\left(\int_{\mathbb{R}} f(v)\mathrm{e}^{-\mathrm{j}\omega(v+u)}\,\mathrm{d}v\right)\mathrm{d}u \\ &= \left(\int_{\mathbb{R}} h(u)\mathrm{e}^{-\mathrm{j}\omega u}\,\mathrm{d}u\right)\left(\int_{\mathbb{R}} f(v)\mathrm{e}^{-\mathrm{j}\omega v}\,\mathrm{d}v\right) \\ &= \widehat{h}(\omega)\widehat{f}(\omega).\end{aligned}$$

□

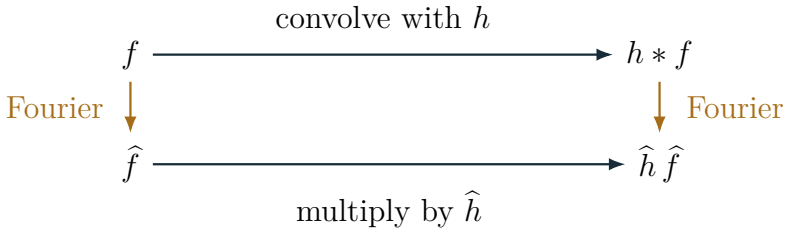


Figure 2.9: The time-domain and frequency-domain descriptions give the same output. The absolute-integrability assumptions above are sufficient conditions, not a claim that they are necessary for every useful filter.

2.10 Fourier, Laplace, and the complex plane

The Fourier transform evaluates a signal against pure oscillations. As in the passage from the DTFT to the z-transform, we can also include an exponential weight. The *two-sided Laplace transform* does this with a complex parameter s:

$$F(s) \coloneqq \int_{\mathbb{R}} f(t)\mathrm{e}^{-st}\,\mathrm{d}t. \tag{2.16}$$

Write $s = \sigma + \mathrm{j}\omega$, with $\sigma, \omega \in \mathbb{R}$. Then

$$\mathrm{e}^{-st} = \mathrm{e}^{-\sigma t}\mathrm{e}^{-\mathrm{j}\omega t}, \qquad F(\sigma + \mathrm{j}\omega) = \widehat{f(\cdot)\mathrm{e}^{-\sigma(\cdot)}}\,(\omega).$$

The real part σ introduces an exponential weight; the imaginary part ω specifies oscillation. Whenever the defining integral converges on the imaginary axis,

$$F(\mathrm{j}\omega) = F(s)\big|_{s=\mathrm{j}\omega} = \widehat{f}(\omega). \tag{2.17}$$

The *region of convergence* is the set of s where the transform exists; for absolute convergence, require $\int_{\mathbb{R}} |f(t)|\mathrm{e}^{-\sigma t}\,\mathrm{d}t < \infty$. A Laplace transform may exist away from the imaginary axis without an ordinary Fourier integral existing on that axis.

For comparison, the z-transform uses

$$X(z) = \sum_{n \in \mathbb{Z}} x[n] z^{-n}, \qquad z = r\mathrm{e}^{\mathrm{j}\Omega}, \quad r > 0,$$

so $z^{-n} = r^{-n}\mathrm{e}^{-\mathrm{j}\Omega n}$. The DTFT is the restriction to $r = 1$, when the series converges there. We use Ω in this comparison for angular frequency per sample.



Figure 2.10: Laplace separates decay and oscillation through Cartesian coordinates (σ, ω). The z-transform separates them through polar coordinates (r, Ω). In both cases, removing the exponential weight leaves a Fourier transform.

Both transforms separate exponential weighting from oscillation. Nevertheless, their domains are different: the Laplace transform acts on functions through an integral, whereas the z-transform acts on sequences through a sum. The imaginary axis and the unit circle are the respective parameter sets on which the exponential weight is removed.

2.11 The Fourier transform of the box is a sinc

We can now return to the moving average and calculate its frequency response. For the centered box $h(t) = \mathbf{1}_{[-1/2,\,1/2]}(t)$, the Fourier integral reduces to integration over an interval. When $\omega \neq 0$, we obtain

$$\begin{aligned} \widehat{h}(\omega) &= \int_{-1/2}^{1/2} \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t \\ &= \left[\frac{\mathrm{e}^{-\mathrm{j}\omega t}}{-\mathrm{j}\omega}\right]_{-1/2}^{1/2} \\ &= \frac{\mathrm{e}^{-\mathrm{j}\omega/2} - \mathrm{e}^{\mathrm{j}\omega/2}}{-\mathrm{j}\omega} \\ &= \frac{-2\mathrm{j}\sin(\omega/2)}{-\mathrm{j}\omega} = \frac{\sin(\omega/2)}{\omega/2}. \end{aligned} \tag{2.18}$$

At $\omega = 0$, compute the integral directly: $\widehat{h}(0) = \int_{-1/2}^{1/2} 1 \, \mathrm{d}t = 1$. With the normalized convention

$$\operatorname{sinc}(v) \coloneqq \begin{cases} \dfrac{\sin(\pi v)}{\pi v}, & v \neq 0, \\ 1, & v = 0, \end{cases}$$

we obtain $\widehat{h}(\omega) = \operatorname{sinc}(\omega/(2\pi))$, not $\operatorname{sinc}(\omega)$.

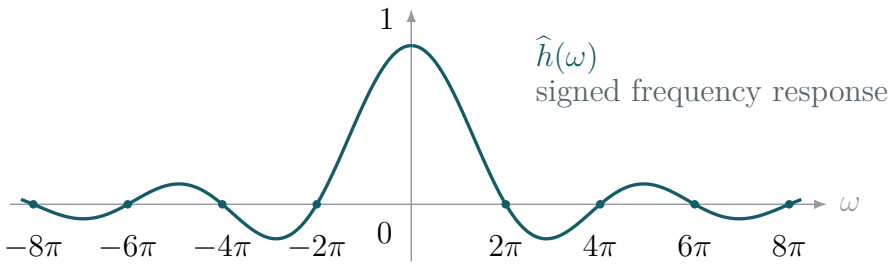


Figure 2.11: The zeros are $\omega = 2\pi k$ for nonzero integers k. The negative sidelobes are negative *values* of the response; the magnitude response is $|\widehat{h}(\omega)|$.

Why is this a low-pass filter? It preserves constants because $\widehat{h}(0) = 1$. Also

$$|\widehat{h}(\omega)| \leq 1, \qquad |\widehat{h}(\omega)| \leq \frac{2}{|\omega|} \quad (\omega \neq 0),$$

by the integral triangle inequality and $|\sin(\omega/2)| \leq 1$, respectively. Rapid oscillations are attenuated. This is *not* an ideal cutoff: sidelobes remain at arbitrarily high frequencies. A negative response changes a cosine's sign, equivalently shifting its phase by π.

2.12 The derivative of an impulse

Differentiation is linear and commutes with translations. We therefore seek an impulse-response description of the differentiator, analogous to the convolution description of a moving average. Since differentiating the input impulse formally gives δ', the required kernel is the derivative of the Dirac impulse. We define this derivative through its action under integration.

Let $\varepsilon > 0$ and form the difference of two shifted impulses:

$$q_\varepsilon(t) \coloneqq \frac{\delta(t+\varepsilon) - \delta(t)}{\varepsilon}. \tag{2.19}$$

The positive impulse is at $-\varepsilon$, because $t + \varepsilon = 0$ there. The negative impulse is at zero. Their total weight is zero.

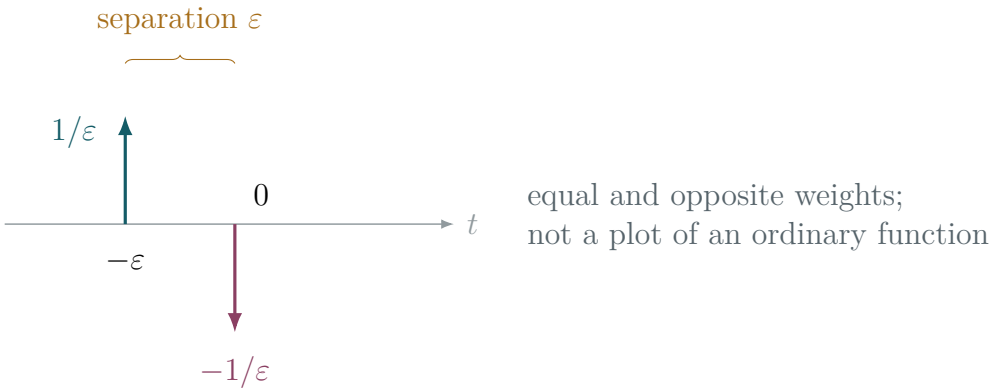


Figure 2.12: A difference quotient of impulses. As the separation shrinks, the weights increase. To understand the limit, first integrate against a differentiable function, then let the separation shrink.

Let φ be differentiable at zero. Applying sifting *before taking a limit* gives

$$\begin{aligned}\int_{\mathbb{R}} \varphi(t) q_\varepsilon(t)\,\mathrm{d}t &= \frac{\varphi(-\varepsilon)-\varphi(0)}{\varepsilon} \\ &= -\frac{\varphi(-\varepsilon)-\varphi(0)}{-\varepsilon} \;\longrightarrow\; -\varphi'(0). \end{aligned} \tag{2.20}$$

We therefore define the derivative impulse by

$$\boxed{\int_{\mathbb{R}} \varphi(t)\delta'(t)\,\mathrm{d}t = -\varphi'(0).} \tag{2.21}$$

Writing $q_\varepsilon \to \delta'$ means that the displayed integrals converge for every φ differentiable at zero. We evaluate each integral first, then take the limit of the resulting numbers; no pointwise limit of impulses is used. The name *doublet* refers to this limiting difference of opposite impulses.

2.13 Integration by parts and differentiation by convolution

The minus sign in (2.21) is the same minus sign that appears in integration by parts. To see the connection, first take ordinary smooth functions r and φ, with r zero outside a finite interval. Integrating the product rule gives

$$\begin{aligned}\int_{\mathbb{R}} r'(t)\varphi(t)\,\mathrm{d}t &= [r(t)\varphi(t)]_{-\infty}^{\infty} - \int_{\mathbb{R}} r(t)\varphi'(t)\,\mathrm{d}t \\ &= -\int_{\mathbb{R}} r(t)\varphi'(t)\,\mathrm{d}t.\end{aligned}$$

The boundary term vanishes because r vanishes outside that interval. For the Dirac impulse, the corresponding derivative rule is

$$\int_{\mathbb{R}} \delta'(t)\varphi(t)\,\mathrm{d}t := -\int_{\mathbb{R}} \delta(t)\varphi'(t)\,\mathrm{d}t = -\varphi'(0).$$

This *definition through integration by parts* agrees with the difference-quotient limit. The result depends only on $\varphi'(0)$, so φ need not vanish outside a finite interval. There is no boundary value of δ to evaluate.

Differentiation by convolution. For differentiable f, fix t and apply (2.21) to $\varphi_t(u) = f(t-u)$. The chain rule gives $\varphi_t'(u) = -f'(t-u)$. Hence

$$\begin{aligned}(\delta' * f)(t) &= \int_{\mathbb{R}} \delta'(u) f(t-u)\,\mathrm{d}u \\ &= -\frac{\mathrm{d}}{\mathrm{d}u} f(t-u)\Big|_{u=0} = -(-f'(t)) = f'(t). \end{aligned} \tag{2.22}$$

There are two sign changes in this calculation. The action of δ' contributes one minus sign, and differentiating the reversed argument $t-u$ contributes the other. Consequently, convolution with δ' gives f', rather than $-f'$.

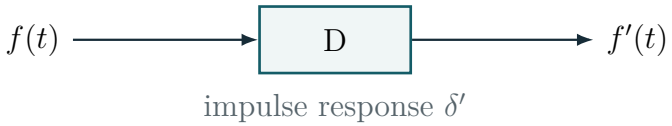


Figure 2.13: Differentiation is convolution with the derivative impulse. Here $(\mathrm{D}f)(t) = f'(t)$.

The same conclusion follows directly from (2.19):

$$(q_\varepsilon * f)(t) = \frac{f(t+\varepsilon) - f(t)}{\varepsilon} \longrightarrow f'(t).$$

Thus the finite-separation two-impulse filter computes a forward difference. As the separation tends to zero, its output tends to the derivative, in agreement with the convolution calculation above.

2.14 Higher derivatives and their frequency responses

For an integer $n \geq 0$, let $\delta^{(n)}$ denote the nth derivative impulse, with $\delta^{(0)} = \delta$. For a function φ with n derivatives near zero, repeated integration by parts gives

$$\int_{\mathbb{R}} \delta^{(n)}(t)\varphi(t)\,\mathrm{d}t = (-1)^n \varphi^{(n)}(0). \tag{2.23}$$

Here is the induction explicitly. The formula for $n = 0$ is sifting. If it holds for n, the derivative rule gives

$$\int \delta^{(n+1)}\varphi = -\int \delta^{(n)}\varphi' = -(-1)^n(\varphi')^{(n)}(0) = (-1)^{n+1}\varphi^{(n+1)}(0).$$

Convolution. For an n-times differentiable input,

$$\begin{aligned}(\delta^{(n)} * f)(t) &= (-1)^n \left.\frac{\mathrm{d}^n}{\mathrm{d}u^n} f(t-u)\right|_{u=0} \\ &= (-1)^n (-1)^n f^{(n)}(t) = f^{(n)}(t). \end{aligned} \tag{2.24}$$

Fourier transform. Apply the same rule to $\varphi(t) = \mathrm{e}^{-\mathrm{j}\omega t}$:

$$\begin{aligned}\widehat{\delta^{(n)}}(\omega) &= (-1)^n \left.\frac{\mathrm{d}^n}{\mathrm{d}t^n} \mathrm{e}^{-\mathrm{j}\omega t}\right|_{t=0} \\ &= (-1)^n (-\mathrm{j}\omega)^n = (\mathrm{j}\omega)^n. \end{aligned} \tag{2.25}$$

As for δ, we evaluate the Fourier integral using the impulse rule with an exponential. It is not an ordinary absolutely convergent integral.

Keep the three identities separate

$$\begin{aligned}\int_{\mathbb{R}} \delta^{(n)}(t)\varphi(t)\,\mathrm{d}t &= (-1)^n \varphi^{(n)}(0), \\ (\delta^{(n)} * f)(t) &= f^{(n)}(t), \\ \widehat{\delta^{(n)}}(\omega) &= (\mathrm{j}\omega)^n. \end{aligned}$$

The sign in the integration identity cancels in convolution through the chain rule.

For $n = 2$, the multiplier is $(\mathrm{j}\omega)^2 = -\omega^2$, exactly as in $\frac{\mathrm{d}^2}{\mathrm{d}t^2}\cos(\omega t) = -\omega^2 \cos(\omega t)$. Unlike the box average, an ideal differentiator amplifies high frequencies: its gain is $|\omega|^n$ for $n \geq 1$.

The box–sinc Fourier pair will reappear in sampling, with the signal and frequency variables exchanging roles. Sinc interpolation can be approached through the signal-processing viewpoint developed here, or through minimum-energy interpolation and projection once the appropriate signal space has been specified. We begin with the first viewpoint: filtering a sampled impulse train to recover its original spectrum.